# Poco — Preserve Originals, Conceal Only

Privacy-first PII anonymisation for LLM workflows, with a fully automated multi-environment DevOps pipeline

**SWE40006 Software Deployment and Evolution**
Group CL01_G01 | Tutorial: Wednesday 10:30
Jamie Soo (104559690) · Timothy Schabel (105334128) · Sahil (105295665)

## The Problem

Every day, people paste contracts, medical notes, and CVs into ChatGPT. Once they do, names, addresses, and financial details are outside their control. Poco makes that safe.

## How Poco Works

Poco detects personal information, replaces each value with a reversible token like [PERSON_1] or [EMAIL_1], and stores the mapping in an encrypted session vault. The redacted text is sent to any LLM, and the response is restored to its original form when it returns.




## Architecture

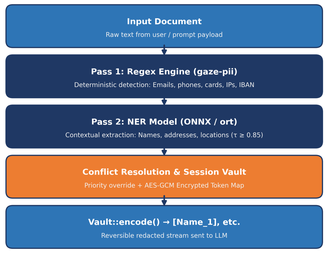



*Figure 1. Two-pass hybrid PII detection engine.*

## DevOps Pipeline

Every code change flows through GitHub Actions, which builds the Rust backend inside the Cargo Lambda container, runs unit, integration, and browser acceptance tests, and deploys to four isolated AWS environments (Development, Testing, Staging, Production). Failed tests block promotion.




## Demo

**Live demo:**
littlepoco.me

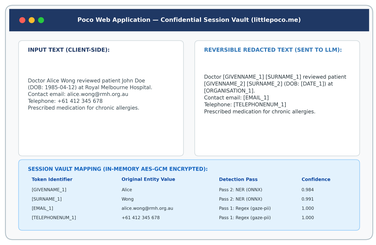



*Figure 2. Poco encode view with coloured token highlighting.*

## Dataset: AI4Privacy OpenPII 1.5M

Analysis of 1,636,375 multilingual documents across 30 languages, 11.9 million PII annotations, and 43 discovered PII entity classes. Full-corpus census completed and cached to Parquet.

| 1,636,375 | 11,887,370 | 43 | 30 |
|---|---|---|---|
| Total examples | Annotations | PII classes | Languages |

### Label Distribution

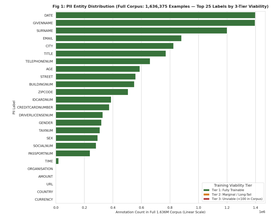



*Figure 3. Top 25 PII labels by viability.*

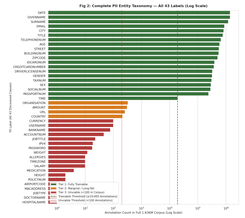



*Figure 4. 43-label taxonomy (log scale).*

### Class Viability

Only 20 of the 43 discovered labels are viable for supervised training. 4 are marginal, and 19 are unviable (<100 corpus annotations, down to single occurrences). The unviable labels are handled by deterministic regex in Poco.

### Language Coverage

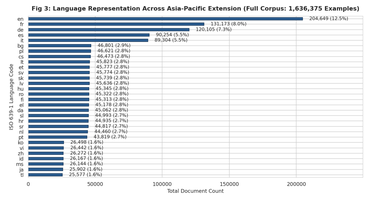



*Figure 5. 30 languages representation.*

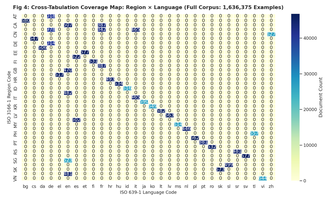



*Figure 6. Region × language mapping.*

### Document Lengths

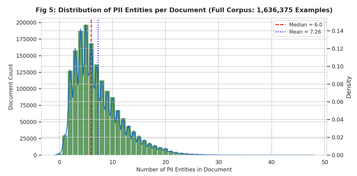



*Figure 7. Entities per document.*

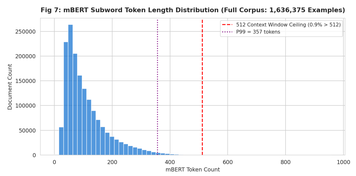



*Figure 8. Token length distribution.*

### Entity Span Lengths & Co-occurrence

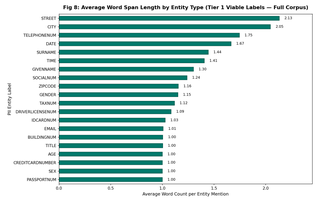



*Figure 9. Word span length by class.*

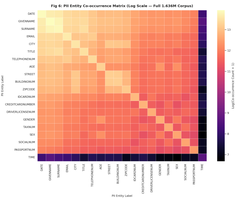



*Figure 9b. Entity co-occurrence matrix.*

## Per-Language Composition

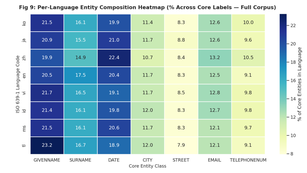



*Figure 10. Per-language entity composition across 8 Asia-Pacific languages.*

Chi-square $p < 10^{-300}$, but Cramér's V = 0.017 shows the effect is negligible — the synthetic generator produces uniform proportions.

## Data Quality & Diagnostic Boxplots

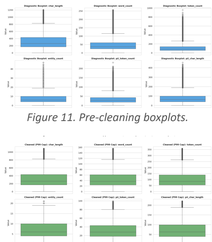


*Figure 11. Pre-cleaning boxplots.*

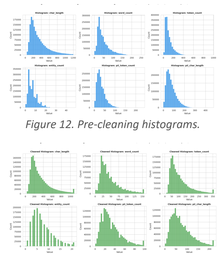


*Figure 12. Pre-cleaning histograms.*



*Figure 13. Winsorised boxplots (P99).*



*Figure 14. Cleaned histograms.*

## Normalisation & Correlation Structure

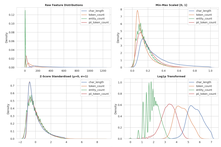


*Figure 15. Transform comparison.*

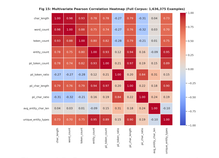


*Figure 16. Pearson correlation matrix.*

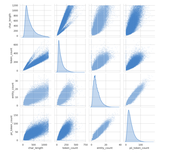


*Figure 17. Bivariate pairplots.*

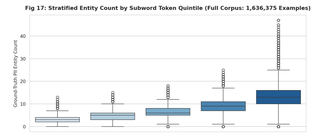



*Figure 18. Stratified entity count.*

## Tokenizer Expansion & Split Parity

Full-corpus census confirmed that all 20 viable classes follow the 80.05% / 19.95% train/val split with $|\Delta| < 0.08\%$.

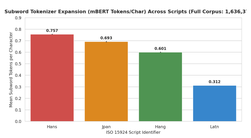



*Fig 19. CJK expansion (2.4×).*

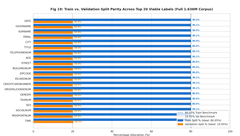



*Fig 20. Split parity.*

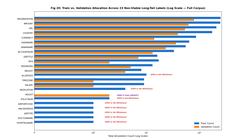



*Fig 21. Rare label imbalance.*

## Key Findings

1. **20 of 43 discovered labels are viable for supervised training**
2. **8 labels have zero validation instances — Vault manual mapping is required**
3. **Chinese text consumes 2.4× more tokens per character than Latin**
4. **Class imbalance is 73:1 between GIVENNAME and TIME**
5. **Perfect 80/20 train/val parity on all viable labels**

**Repository:**
github.com/Schnitze1/SWE40006-Group-Project

**Full EDA Report:**
woojin1069/pii-masking-openpii-1.5m